

Customer Churning Strategy

Customer Churn Prediction, Segmentation & Retention Recommendation System

1. Project Objective

Build an end-to-end **Customer Churning Strategy System** that identifies customers who are likely to churn, understands the behavioural characteristics of different customer segments, and recommends an appropriate retention action.

The system should not stop at:

"Which customers are likely to churn?"

It should progress toward:

"Why might the customer churn, which customers should be targeted, and what retention reward should be offered?"

The project should demonstrate the complete reasoning flow:

Business Problem → Data Understanding → ML Approach → Model Selection → Technology Selection → Architecture → Implementation → API Integration → Retention Action → Business Outcome

2. Business Problem

Customer churn directly affects recurring revenue and customer lifetime value. A company may have thousands or millions of customers, but it cannot manually identify which customers are at risk of leaving.

The business wants to build a system that can:

1. Identify customers with a high probability of churn.
2. Understand the factors associated with churn.
3. Identify different customer behaviour segments.
4. Prioritize customers for retention campaigns.
5. Recommend an appropriate reward or intervention.

6. Expose the prediction through an API so that other business applications can consume it.
7. Provide a simple interface for business users to analyse individual or groups of customers.

Core Business Question

How can machine learning help the company proactively identify customers at risk of churning and recommend targeted retention actions before the customer leaves?

3. Dataset

Use the following Kaggle dataset:

Customer Churn Prediction Dataset – 1M

[Kaggle Dataset](#)

A published analysis of this dataset reports approximately **1,000,000 customer records and 32 columns**, covering customer demographics, account/service information, billing and behavioural characteristics, with churn as the target variable. The exact schema should be confirmed by the candidates after downloading the dataset rather than assumed from a secondary description.

Candidate Responsibility

Before modelling, candidates must document:

- Number of rows and columns
- Data types
- Target variable
- Numerical features
- Categorical features
- Missing values
- Duplicate records
- Outliers
- Class distribution
- Potential data leakage
- Highly correlated variables
- Features that could be useful for explaining churn

4. Context and Data Understanding

Candidates should first understand **what the data represents from a business perspective**.

The analysis should answer questions such as:

Customer Profile

- Who are the customers?
- What customer characteristics are available?
- What services/products do they use?
- How long have they been customers?

Behaviour

- How frequently are services being used?
- Are there indicators of declining engagement?
- Are there complaints or service issues?
- Are there payment-related behavioural patterns?

Financial

- What are the customer's charges?
- How does customer value vary?
- Are high-value customers also at risk of churn?

Churn

- What percentage of customers churn?
- Which variables appear associated with churn?
- Are there particular customer segments with higher churn?

The candidates should distinguish between:

Correlation / association

and

causal evidence.

The dataset can support predictive analysis, but it does not automatically prove that a particular intervention will prevent churn.

5. Possible ML Approaches

The project should evaluate multiple approaches rather than selecting one algorithm without justification.

A. Supervised Classification

Objective:

Predict whether a customer is likely to churn.

Possible models:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. SVM
5. XGBoost / Gradient Boosting
6. Neural Network / ANN using PyTorch

The target could be represented as:

Churn = 1 → Customer likely to churn

Churn = 0 → Customer likely to stay

The output should include a **churn probability**, not only a Yes/No prediction.

Example:

Customer ID: C10291

Churn Probability: 0.82

Risk Category: High

6. Required ML Models

The following models must be investigated as part of the project.

6.1 Decision Tree

Why use it?

Decision Trees are useful because they provide interpretable decision rules.

Example:

IF tenure < X
AND monthly charges > Y
AND complaints > Z
THEN high churn risk

Project Usage

Use the Decision Tree to:

- Build an interpretable baseline.
 - Understand important decision paths.
 - Explain churn risk to business users.
 - Identify important features.
-

6.2 Support Vector Machine — SVM

Why use it?

SVM can be useful for classification when the classes are separated by complex decision boundaries.

Project Usage

Use SVM as a classification model and compare it against:

- Decision Tree
- Logistic Regression
- Random Forest
- Gradient Boosting

Candidates should investigate the impact of:

- Kernel

- C
- Gamma
- Feature scaling

Because the dataset contains approximately one million records, candidates must also discuss the **computational implications of training SVM at this scale** rather than blindly applying it to the complete dataset.

7. Clustering

Churn prediction answers:

Who is likely to churn?

Clustering answers:

What types of customers exist?

Candidates should use clustering to identify behavioural/customer segments.

Possible features:

- Tenure
- Monthly charges
- Total charges
- Usage
- Number of services
- Complaints
- Customer satisfaction
- Payment behaviour

Example output:

Segment 1 → High-value, highly engaged customers
Segment 2 → Low-value, low-engagement customers
Segment 3 → High-charge, low-engagement customers
Segment 4 → New customers with increasing usage

These segments can subsequently be combined with churn risk.

8. PCA — Principal Component Analysis

PCA should be used primarily for **dimensionality reduction and analysis**, not as a replacement for the churn model.

Candidates should investigate:

- Feature variance
- Principal components
- Explained variance ratio
- 2D/3D visualisation
- Impact of PCA on model performance

Example:

Original Features



Standardization



PCA



Principal Components



Visualisation / Model

PCA can also help visualize whether churn/non-churn customers or customer segments exhibit separable patterns.

9. DBSCAN

DBSCAN should be used for **density-based customer segmentation and anomaly detection**.

Unlike K-Means, DBSCAN does not require the number of clusters to be specified beforehand and can identify noise/outlier observations.

Candidates should investigate:

- `eps`
- `min_samples`

- Cluster distribution
- Noise points
- Customer characteristics of each cluster

Potential business use:

Identify unusual customer behaviour that does not belong to a normal customer segment.

For example, a customer could have:

- unusually high complaints,
 - unusually high charges,
 - unusually low usage,
 - or an unusual combination of behavioural attributes.
-

10. Additional Models to Include

The project should also include the following models.

10.1 Logistic Regression — Baseline

Use as the primary baseline model.

Advantages:

- Simple
 - Fast
 - Interpretable
 - Provides probability estimates
 - Useful benchmark for more complex models
-

10.2 Random Forest

Use as a stronger tree-based baseline.

Useful for:

- Non-linear relationships

- Feature importance
 - Robust classification
 - Comparing ensemble learning against a single Decision Tree
-

10.3 Gradient Boosting / XGBoost

Use as a high-performance tabular-data model.

Candidates should compare its performance with Random Forest and Decision Tree.

Potential metrics:

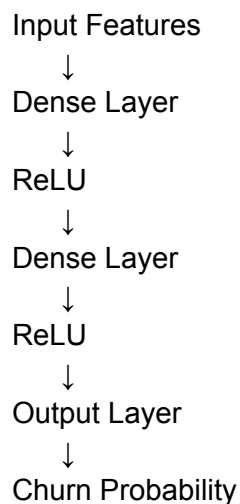
- ROC-AUC
- PR-AUC
- Precision
- Recall
- F1
- Confusion Matrix

For churn, candidates must explain why **accuracy alone is insufficient**.

10.4 Neural Network / ANN — PyTorch

Build a simple feed-forward neural network using PyTorch.

Example architecture:



Candidates should compare the ANN against classical ML models and explain whether the additional complexity is justified.

11. Model Evaluation Strategy

The candidates should not select a model simply because it has the highest accuracy.

The evaluation should consider:

Metric	Business Meaning
Precision	How many predicted churners actually churn?
Recall	How many actual churners were identified?
F1	Balance between precision and recall
ROC-AUC	Overall ranking ability
PR-AUC	Useful when churn is relatively imbalanced
Confusion Matrix	Understand prediction errors
Inference Time	Production responsiveness
Model Size	Deployment considerations
Interpretability	Ability to explain recommendations

The final model selection should be justified based on **business requirements + model performance + operational cost**, not accuracy alone.

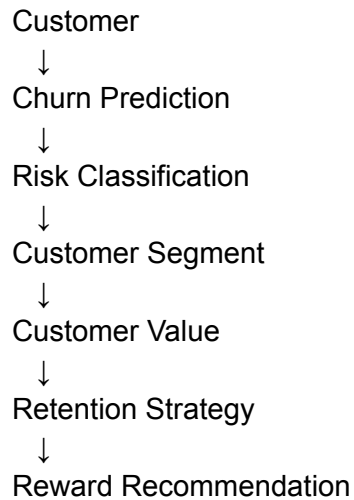
12. From Churn Prediction to Churn Strategy

This is the most important part of the project.

A conventional churn model produces:

Customer → Churn Probability

The proposed system should extend this to:



Example:

Churn Risk	Customer Value	Recommended Action
Low	Low	No intervention
Low	High	Loyalty benefit
Medium	High	Personalized reward
High	Low	Low-cost retention offer
High	High	Priority retention campaign

13. FastAPI Use Case — Real-Time Retention Recommendation

FastAPI should expose the ML model as a production-style prediction service.

Use Case

A CRM/application sends customer information to the API.

Example:

POST /predict-churn

Input:

```
{
  "customer_id": "C10291",
  "tenure": 14,
  "monthly_charges": 89.50,
  "total_charges": 1253.00,
  "complaints": 3,
  "service_calls": 5
}
```

The API returns:

```
{
  "customer_id": "C10291",
  "churn_probability": 0.82,
  "risk_level": "HIGH",
  "customer_segment": "HIGH_VALUE_AT_RISK",
  "recommended_action": "RETENTION_REWARD",
  "reward": "20% discount for 3 months"
}
```

14. Reward Recommendation Logic

The API should not simply return:

Churn = Yes

It should support a **retention decision layer**.

Example:

IF churn_probability >= 0.75
AND customer_value = HIGH
→ Premium retention reward

IF churn_probability >= 0.60
AND customer_value = MEDIUM
→ Moderate discount / loyalty reward

IF churn_probability < 0.40
→ No retention reward

IF customer is highly engaged
AND churn probability is low
→ Loyalty benefit rather than discount

The reward logic should be configurable rather than hard-coded throughout the application.

15. Important Extension — Uplift / Treatment Thinking

The project should explicitly distinguish:

Churn Prediction

Who is likely to churn?

from:

Retention Optimization

Who is likely to change their behaviour because of a retention intervention?

This distinction is important because a high-risk customer is not automatically the best customer to target with an expensive reward.

Uplift modelling estimates the incremental effect of an intervention and is therefore more closely aligned with retention decision-making. Research on customer retention has specifically examined uplift models for targeting customers based on the expected effect of retention campaigns rather than churn probability alone.

Dataset Limitation

If the Kaggle dataset does not contain historical treatment/intervention information such as:

Customer

- Received Reward?
- Reward Type
- Reward Cost
- Stayed/Churned

then candidates **must not claim to have built a validated causal uplift model**.

Instead, they should:

1. Build the churn prediction system.
 2. Build a rule-based reward recommendation layer.
 3. Explain what additional treatment/control data would be required for true uplift modelling.
 4. Optionally simulate a treatment dataset for demonstration, clearly labelling it as synthetic.
-

16. Technology Selection

Required Technology Stack

Programming

- Python

Machine Learning

- Scikit-learn
- PyTorch

API

- FastAPI
- Pydantic

Frontend

Choose one:

- Streamlit
- Gradio

Supporting Libraries

Candidates may use:

- Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Plotly
 - Joblib
 - Uvicorn
-

17. Why These Technologies?

Python

Used as the primary development language because the complete ML pipeline can be developed within one ecosystem.

Scikit-learn

Used for:

- Preprocessing
- Train/test split
- Decision Tree
- SVM
- Logistic Regression
- Random Forest
- Clustering
- PCA
- DBSCAN
- Model evaluation

PyTorch

Used for:

- ANN implementation
- Training loop
- Loss calculation
- Optimisation
- Neural-network experimentation

FastAPI

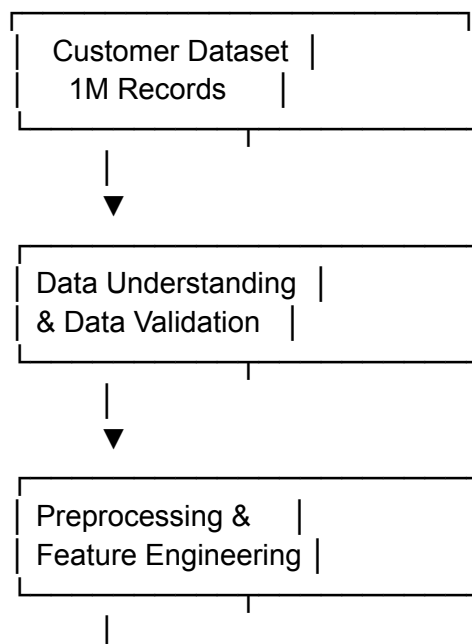
Used to convert the trained ML pipeline into a reusable prediction service.

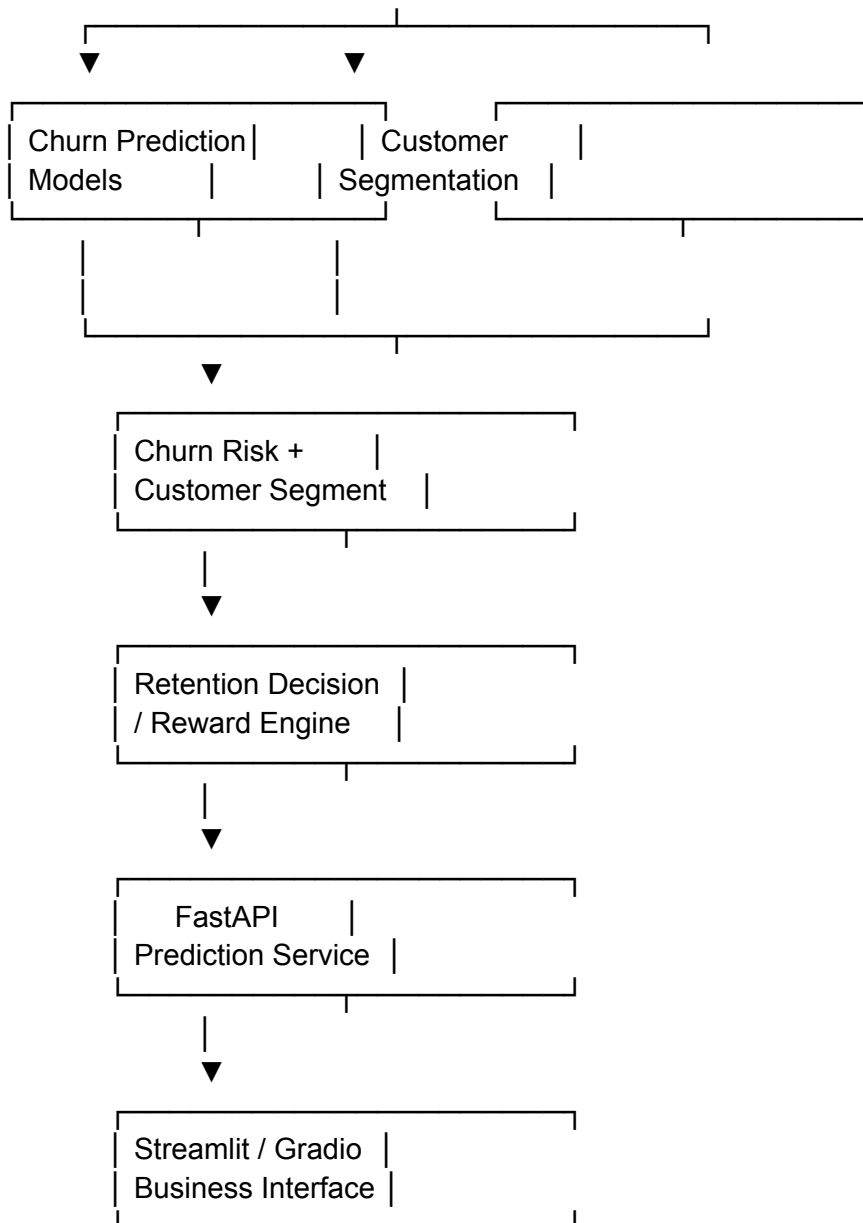
Streamlit / Gradio

Used to provide a simple business-facing interface for:

- Customer-level prediction
 - Churn probability
 - Risk category
 - Segment
 - Recommended reward
 - Model insights
-

18. Proposed Architecture





19. Implementation Flow

Phase 1 — Data Understanding

- Download dataset.
- Inspect schema.
- Validate data.

- Analyse missing values.
- Analyse duplicates.
- Understand target distribution.
- Perform EDA.

Deliverable

EDA notebook + data dictionary.

Phase 2 — Data Preparation

Implement:

- Missing-value treatment
- Categorical encoding
- Numerical scaling
- Outlier analysis
- Feature selection
- Train/validation/test split

Candidates should avoid data leakage during preprocessing.

Phase 3 — ML Experiments

Build and compare:

Classification

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. SVM
5. XGBoost/Gradient Boosting
6. PyTorch ANN

Unsupervised Learning

7. K-Means
8. DBSCAN
9. PCA

The purpose of each model must be documented.

20. Model Selection Report

Candidates should produce a comparison such as:

Model	ROC-AUC	PR-AUC	Recal I	F1	Training Time	Inference Time
Logistic Regression						
Decision Tree						
Random Forest						
SVM						
XGBoost						
ANN						

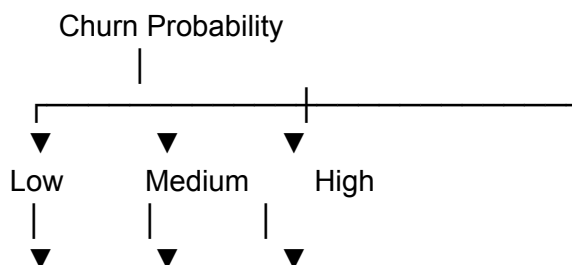
The candidate must explain:

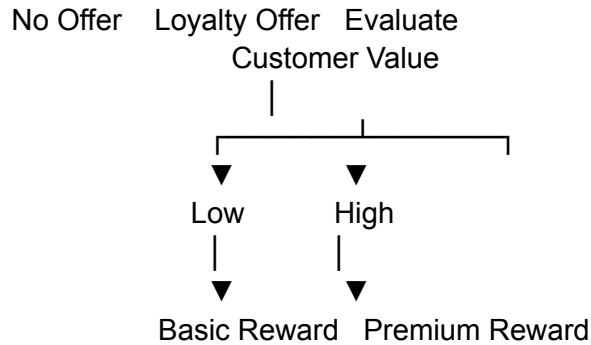
Why was the final model selected for production?

21. Retention Strategy Engine

The final system should convert model output into a business action.

Example:





Possible rewards:

- Discount
- Free upgrade
- Additional service
- Loyalty points
- Extended subscription
- Free support/service benefit
- Personalized offer

The project should also consider **reward cost** so that the system does not recommend expensive rewards to every high-risk customer.

22. FastAPI Endpoints

Minimum endpoints:

Health Check

GET /health

Churn Prediction

POST /predict-churn

Customer Segment

POST /segment-customer

Retention Recommendation

POST /recommend-retention

Combined Endpoint

POST /customer-strategy

Example:

```
{
  "customer_id": "C10291",
  "churn_probability": 0.82,
  "risk_level": "HIGH",
  "segment": "HIGH_VALUE_AT_RISK",
  "recommended_reward": "20% discount for 3 months",
  "reason": [
    "Low tenure",
    "High monthly charges",
    "High support interactions"
  ]
}
```

23. Frontend Requirements

Build the frontend using **Streamlit** or **Gradio**.

The interface should contain:

Customer Prediction

Input customer attributes and display:

Churn Probability

Risk Level

Customer Segment

Recommended Action

Recommended Reward

Customer Analytics

Display:

- Churn distribution
- Segment distribution
- High-risk customer count
- Feature importance
- Model performance

Retention Dashboard

Show:

- High-risk customers
 - High-value/high-risk customers
 - Suggested rewards
 - Estimated reward cost
 - Potential retention opportunity
-

24. Cost, Efficiency and Scalability

Candidates must think beyond model accuracy.

Dataset Scale

With approximately one million records, the project should consider:

- Memory usage
- Training time
- Feature preprocessing time
- Model inference latency
- Batch vs real-time prediction

The SVM experiment in particular should include a discussion of computational scalability.

Batch Prediction

Suitable for:

Daily / weekly churn scoring

Example:

1M customers



Batch ML prediction



High-risk customer list



Retention campaign

Real-Time Prediction

Suitable for:

Customer opens application



API request



Churn prediction



Retention recommendation



Personalized reward

25. Business Outcome

The final system should define measurable KPIs.

Primary KPIs

- Churn rate
- Customer retention rate
- Number of customers retained
- Revenue retained
- Customer Lifetime Value
- Retention campaign ROI

ML KPIs

- Recall of churners
- Precision
- ROC-AUC
- PR-AUC
- Prediction latency

Operational KPIs

- API response time
 - Batch processing time
 - Model size
 - Infrastructure cost
 - Number of customers scored per batch
-

26. Business Impact Calculation

Candidates should create a simple financial simulation.

For example:

Customers targeted
×
Expected incremental retention
×
Average customer value
=
Potential retained value

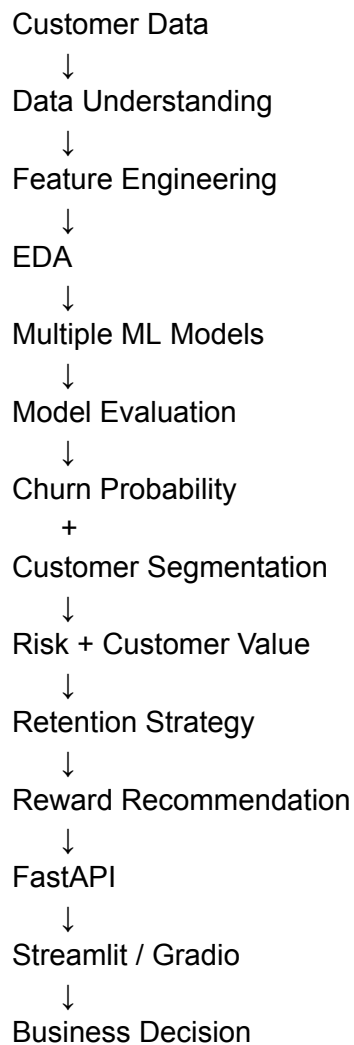
Then:

Potential retained value
-
Retention reward cost
-
Campaign cost
=
Estimated net business impact

This is preferable to claiming that a model is successful solely because it achieved a high F1 or ROC-AUC.

27. Final Expected Solution

The completed project should demonstrate:



The final solution should therefore be positioned as a:

Customer Churn Prediction and Retention Strategy Platform

rather than simply a **Customer Churn Prediction Model**.

28. Expected Candidate Deliverables

1. Data Understanding & EDA Notebook
 2. Feature Engineering Pipeline
 3. ML Training Notebook
 4. Model Comparison Report
 5. Clustering + PCA + DBSCAN Analysis
 6. Final Trained Model
 7. FastAPI Application
 8. Retention Recommendation Engine
 9. Streamlit/Gradio Frontend
 10. API Documentation
 11. Architecture Diagram
 12. Business Impact / ROI Analysis
 13. README with Setup Instructions
 14. Final Presentation/Demo
-

29. Key Reasoning Expected from Candidates

The candidates should be able to answer the following during the project review:

Business

- Why does the business care about churn?
- What is the financial impact of churn?
- Why shouldn't every customer receive a reward?

Data

- Which features are relevant to churn?
- Is the dataset balanced?
- Are there leakage risks?
- What behavioural patterns are visible?

ML

- Why Decision Tree?
- Why SVM?
- Why Random Forest?
- Why XGBoost?
- Why ANN?
- Why clustering?
- Why PCA?
- Why DBSCAN?
- Which model should go into production and why?

Engineering

- Why FastAPI?
- Why Streamlit/Gradio?
- How will the model be loaded?
- How will preprocessing be kept consistent between training and inference?
- How will one million customers be scored efficiently?
- What happens if the model needs to be retrained?

Business Strategy

- Why should a high-risk customer receive a reward?
- How do we account for reward cost?
- How do we identify high-value customers?
- How do we measure whether the reward actually prevented churn?
- What additional data would be required to build a genuine uplift model?

30. Success Criteria

The project is considered complete when the candidate can demonstrate an end-to-end flow:

**Customer → Churn Risk → Segment → Business Reason → Retention Action → Reward
→ API → User Interface**

The final objective is not merely to build a model with good predictive performance, but to demonstrate how an ML system can be integrated into a **real customer-retention decision workflow**.